Analytics
Regression Analysis

CSDA 5110

Regression

- Regression Analysis is used to examine the relationship among quantitative variables.
- The technique is used to predict the value of one variable (the dependent variable - y)based on the value of other variables (independent variables x_1, x_2,…x_k.)

Relations between variables

- Regression: relationship between variables
- Functional: $Y = f(X)$
- Statistical Relationship
  - Variation

Regression Models and their uses

- Y varies with X in a systematic way
- Scattering of points around the curve of statistical relationship
  - CGPA and HSGPA
- Model Postulates
  - Probability distribution of Y for each level of X
  - The means of these probability distributions vary in some systematic fashion with X
- Uses
  - Prediction
  - Variable screening
  - Model specification
  - Parameter estimation

Regression Models and their uses

Steps

1) Variable Selection
2) Functional Form: Linear or Non-Linear
3) Scope of the Model
4) Residual analysis
5) Estimation
6) Quality of model

Note

- Statistically significant association between Y and set of X’s does not necessarily implies causality
- Regression analysis only allow one to evaluate a hypothesized relationship
- Limitation – Garbage in Garbage out

The Model

- The first order linear model

$$y = \beta_0 + \beta_1 x + \varepsilon$$

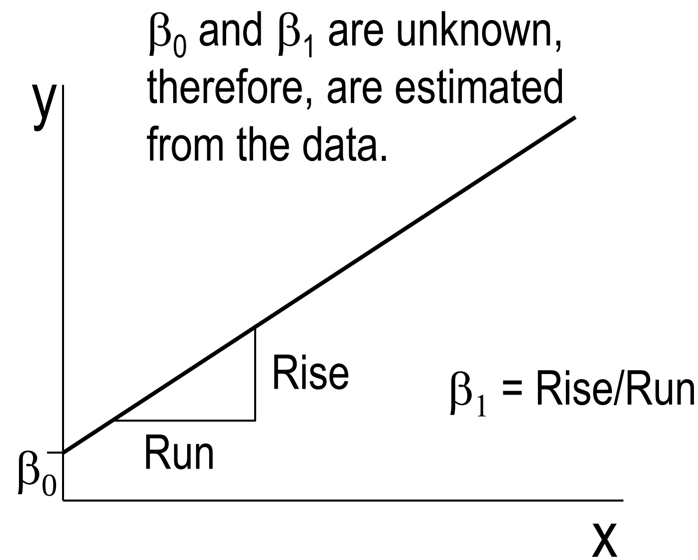
y = dependent variable

x = independent variable

β_0 = y-intercept

β_1 = slope of the line

ε = error variable

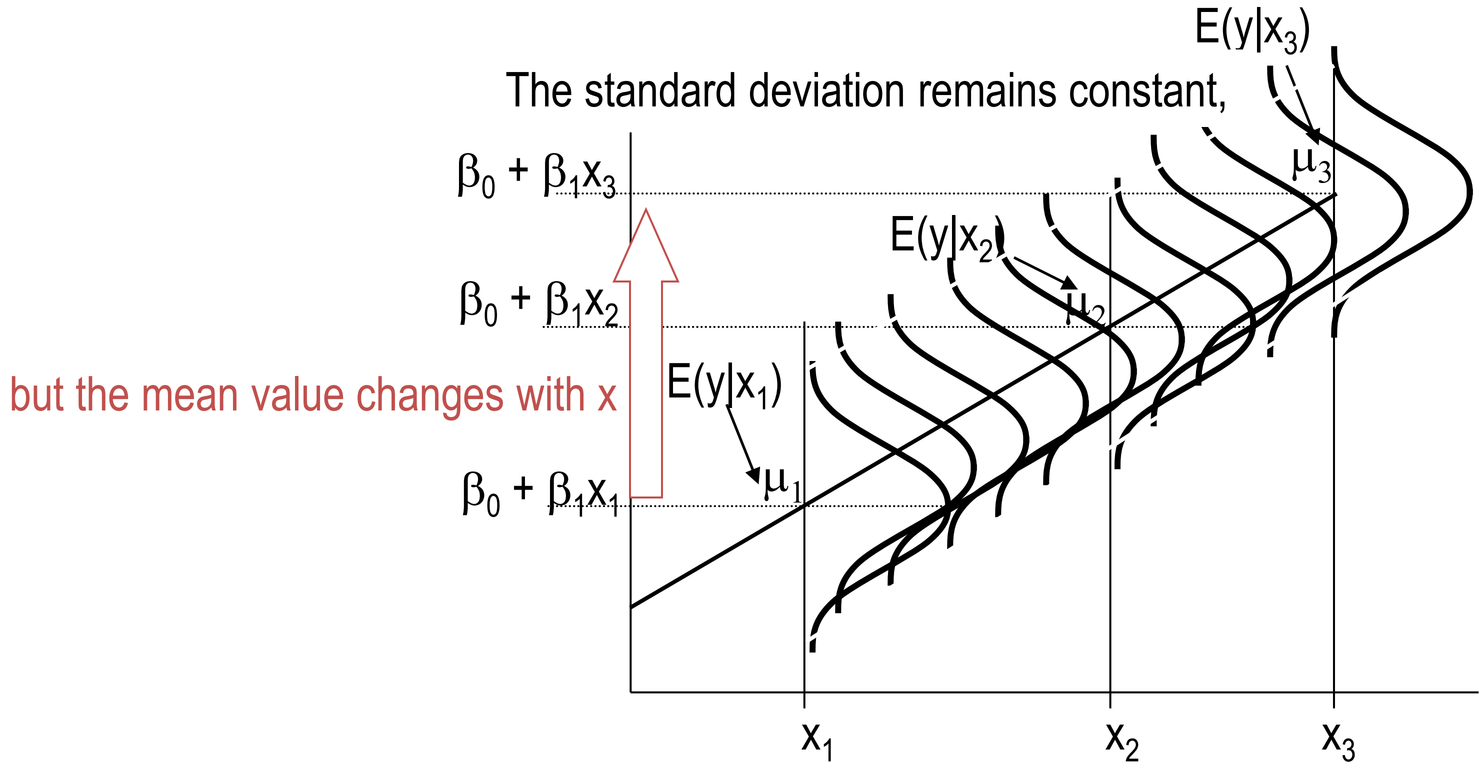


Assumptions

- The error ε is a critical part of the regression model.
- Four requirements involving the distribution of ε must be satisfied.
  - The probability distribution of ε is normal.
  - The mean of ε is zero: $E(\varepsilon) = 0$.
  - The standard deviation of ε is σ_ε for all values of x.
  - The set of errors associated with different values of y are all independent.
- x_i ‘s are nonrandom, observed with negligible error

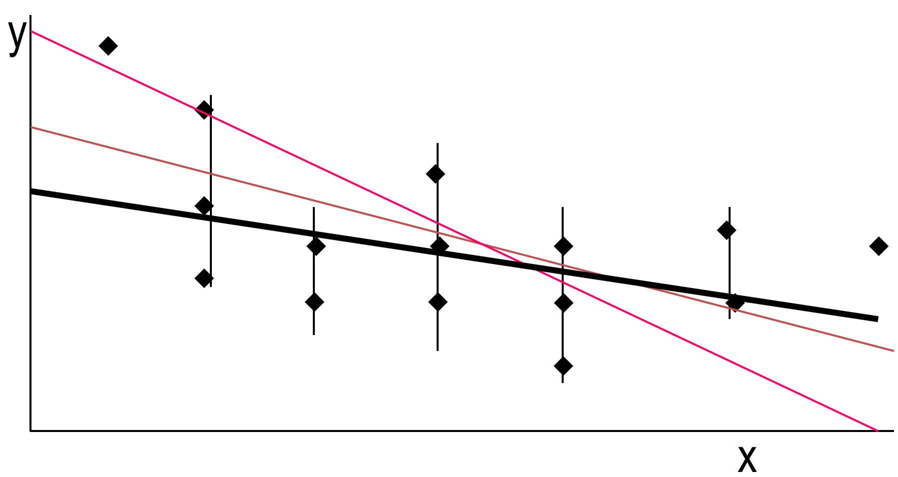
From the first three assumptions we have:

1. y is normally distributed with mean
2. $E(y) = b_0 + b_1x$, and a constant standard deviation σ_e
3. Variability of y remains same across all X values
4. Responses are Independent



Estimating the Coefficients

- The estimates are determined by
  - drawing a sample from the population of interest,
  - calculating sample statistics.
  - producing a straight line that cuts into the data.



The question is:
Which straight line fits best?

Estimating the Coefficients

- Question
  - How do you know that E(Y|X=x_i) and x_i ae linearly related?
    - Hypothesize a linear relationship based on scientific knowledge
    - Scatterplot
  - Consider fitting SLR model in a restricted range of x-values
  - Consider linearizing the model
  - Use non-linear modeling

Assessing the Model

- The least squares method will produce a regression line whether or not there is a linear relationship between x and y.
- Consequently, it is important to assess how well the linear model fits the data.
- Several methods are used to assess the model:
  - Testing and/or estimating the coefficients.
  - Using descriptive measurements.

The best line is the one that minimizes the sum of squared vertical differences between the points and the line.

To calculate the estimates of the coefficients that minimize the differences between the data points and the line, use the formulas:

$$b_1 = \frac{\text{cov}(X,Y)}{s_x^2}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

The regression equation that estimates the equation of the first order linear model is:

$$\hat{y} = b_0 + b_1 x$$

Making Statistical Inference about β_1

H_0: $\beta_1 = 0$

H_1: $\beta_1 = 0$ (or < 0,or > 0)

- Test Statistic

- Decision Rule

- Confidence Interval

Example 1

```
output.lm <- lm(workhrs ~ lotsize,
Toluca_Example1)
output.lm
summary(output.lm)

Call:
lm(formula = workhrs ~ lotsize, data = Toluca_Chapter1)

Residuals:
   Min      1Q  Median      3Q     Max
-83.876 -34.088  -5.982  38.826 103.528

Coefficients:
            Estimate Std. Error  t value   Pr(>|t|)
(Intercept)   62.366     26.177   2.382     0.0259 *
lotsize        3.570      0.347  10.290   4.45e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.82 on 23 degrees of freedom
Multiple R-squared:  0.8215,	Adjusted R-squared:  0.8138
F-statistic: 105.9 on 1 and 23 DF,  p-value: 4.449e-10
```

Confidence Intervals on Mean Response

Prediction Intervals of new value at X = x_0

- An interval where 100(1-α)% of the future observations are expected to fall.
- X within range (Interpolation)
- X outside the range (extrapolation)
- $\hat{Y}_{x_0}$ is estimate as well as predicted value if x_0 is future value
- Variation in possible location of the distribution of Y
- Variation within the probability distribution of Y

Example 1

```
predict(output.lm, interval="confidence",level=0.95)
predict(output.lm,interval="prediction",level=0.95)
```

```
predict(output.lm,newdata=data.frame(lotsize=c(35,45,50,65,100,110)),interval="confidence",level=0.95)

predict(output.lm,newdata=data.frame(lotsize=c(35,45,50,65,100,110)),interval="prediction",level=0.95)
```

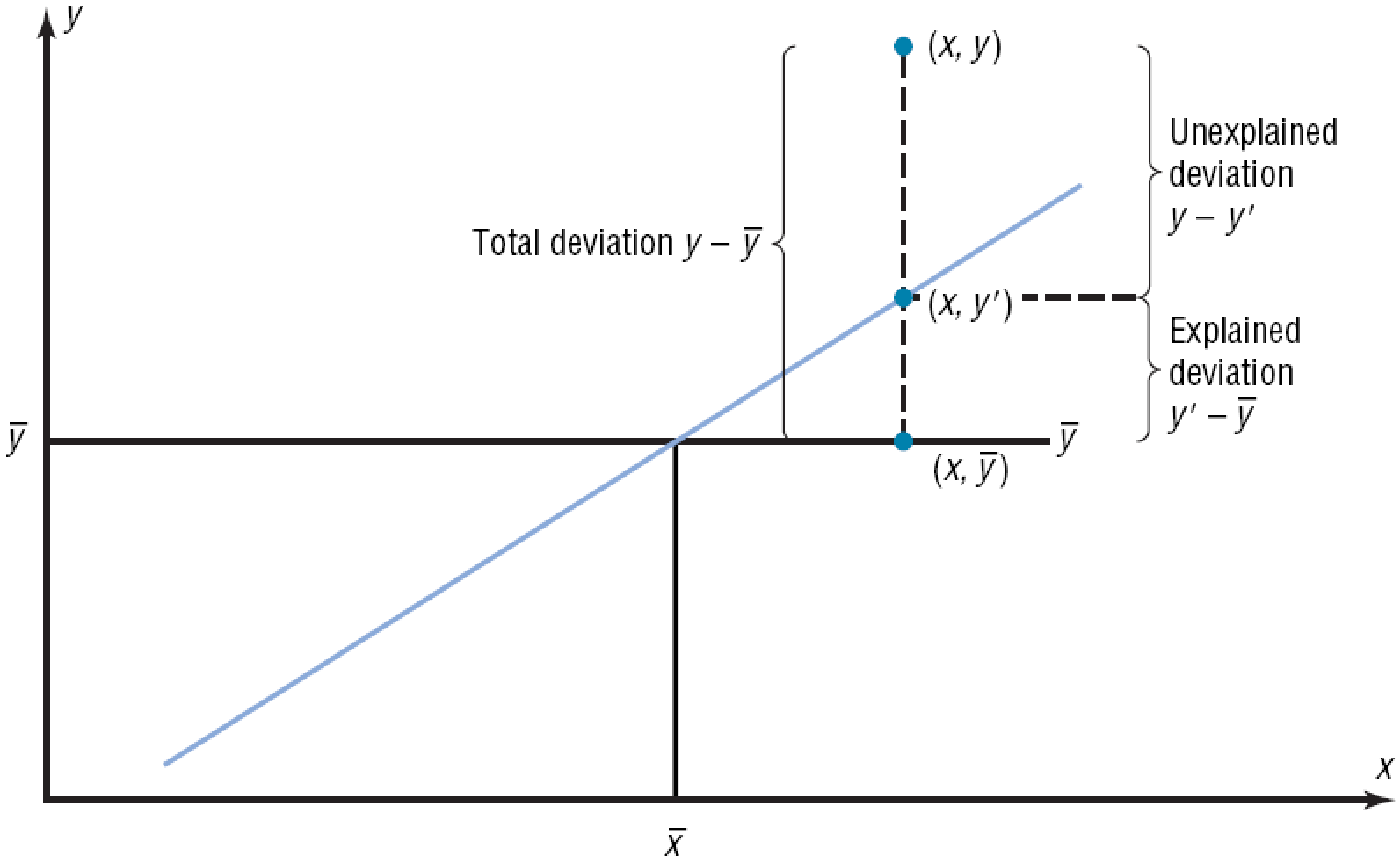
	fit	lwr	upr
1	187.3229	155.0873	219.5586
2	223.0249	196.0060	250.0439
3	240.8760	216.0948	265.6571
4	294.4290	273.9129	314.9451
5	419.3861	389.8615	448.9106
6	455.0881	419.9835	490.1927

	fit	lwr	upr
1	187.3229	81.30465	293.3412
2	223.0249	118.47466	327.5752
3	240.8760	136.88151	344.8704
4	294.4290	191.36760	397.4904
5	419.3861	314.16040	524.6117
6	455.0881	348.16254	562.0136

Partitioning the total variability in SLR

- The **total variation** $\sum(y-\bar{y})^2$ is the sum of the squares of the vertical distances each point is from the mean.
- The total variation can be divided into two parts: that which is attributed to the relationship of *x* and *y*, and that which is due to chance (variability due to error).

Variation



y
(x, y)
Unexplained deviation $y - y'$
Total deviation $y - \bar{y}$
(x, y')
Explained deviation $y' - \bar{y}$
$\bar{y}$
$\bar{y}$
$(x, \bar{y})$
x
$\bar{x}$

$$\sum(y_i - \bar{y})^2 = \sum(\hat{y}_i - \bar{y})^2 + \sum(y_i - \hat{y}_i)^2$$

Sum Square Total (SST): $\sum(y_i - \bar{y})^2 = S_{yy}$

Sum Square Regression (SSR): $\sum(\hat{y}_i - \bar{y})^2 = \frac{{S_{xy}}^2}{S_{xx}}$

Sum Square Error (SSE): $\sum(y_i - \hat{y}_i)^2 = S_{yy} - \frac{{S_{xy}}^2}{S_{xx}}$

ANOVA Table

Source of Variation	Degrees of Freedom (DF)	Sum Squares (SS)	Mean Squares Error (MSE	F
Regression	1	SSR $=\sum(\hat{y}_i - \bar{y})^2$	MSR =SSR/1	MSR/MSE
Error	n-2	SSE = $\sum(y_i - \hat{y}_i)^2$	MSE= SSE/n-2	
Total	n-1	SST = $\sum(y_i - \bar{y})^2$		

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output.lm <- lm(workhrs ~ lotsize,
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output.lm
summary(output.lm)
anova(output.lm)
```

```
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```

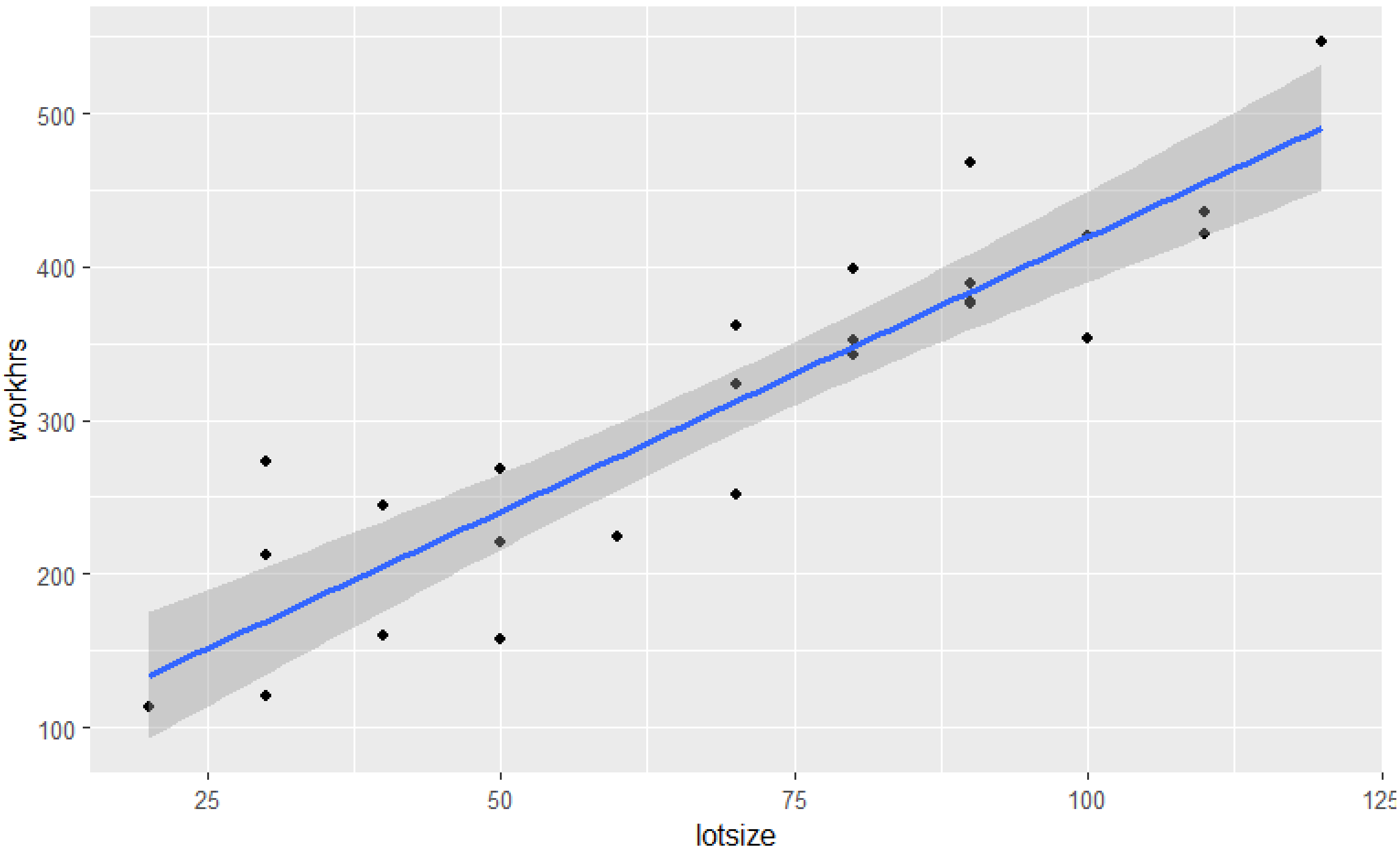
```
Analysis of Variance Table

Response: workhrs
          Df Sum Sq Mean Sq F value    Pr(>F)
lotsize    1 252378  252378  105.88 4.449e-10 ***
Residuals 23  54825    2384
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Quality of the fitted Model

- Measures association between X and Y
- The **coefficient of determination** is the ratio of the explained variation to the total variation.
  - $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- Proportion of variability in Y that can be explained by the regression model

Quality of the fitted Model



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Quality of the fitted Model

- Another way to arrive at the value for r^2 is to square the correlation coefficient.
- *Interpretation*
  - $0 \leq R^2 \leq 1$
  - R^2 *should be high. Accepatble value depends on application*
  - *Throwing in more variables may increase* R^2 *but may result in other problems like multicollinearity making prediction unstable*
  - R^2 *is not a reliable estimator of* ρ^2 *when sample size is small*
  - R^2 *can be high*
    - *Slope of regression is large*
    - *Spread of the regressor data is large*

Diagnostics and Remedial Measures in SLR

- Residuals

$e_i = y_i$-$\hat{y}_i$

e_i = y_i-b_0- b_1x

Error

ϵ_i = Y_i-$E(Y)$

Residuals are used to evaluate the validity o the model assumptions.

$$\epsilon_i \sim N(0, \sigma^2)$$

Mean: $\bar{e} = \frac{\sum e_i}{n} = 0$

Variance: $\frac{\sum(e_i - \bar{e})^2}{n-2} = \frac{\sum(e_i)^2}{n-2} = \frac{SSE}{n-2} = MSE$

Non-Independence: e_i are not independent random variable (r.v.) because they involve fitted values $\hat{y}_i$ which are based on the same fitted regression function.

Two constraints: $\sum e_i$=0 ; $\sum x_i e_i$=0 (large samples -> can ignore dependency)

Departures from Model Studied by Residuals

- Regression function is not linear
- Constant variance
- Independence of error terms
- Outliers (model fits all but a few)
- Non-normality
- One or several important predictor variables have been omitted

Diagnostic Plots of Residual

- Non-Linearity
  - Scatter plot
  - Residual plot against predictor variable
  - Residual plot against fitted values
  -

Diagnostic Plots of Residual

- Non-Constancy of error variance
  - Residual plot against predictor variable
  - Residual plot against fitted values
  - Plot of absolute values of the residuals against predictor or fitted values (more useful when few data values)
  - Plot of squared residuals against predictor or fitted values

Diagnostic Plots of Residual

- Presence of outliers
  - Boxplot of residuals
  - Residual plot against predictor variable
  - Residual plot against fitted values
  - Plotting semi-studentized residuals (4 standard deviations -> outliers)

Diagnostic Plots of Residual

- Non-Independence
  - Cyclic
  - Autocorrelation (Durbin-Watson test)

Diagnostic Plots of Residual

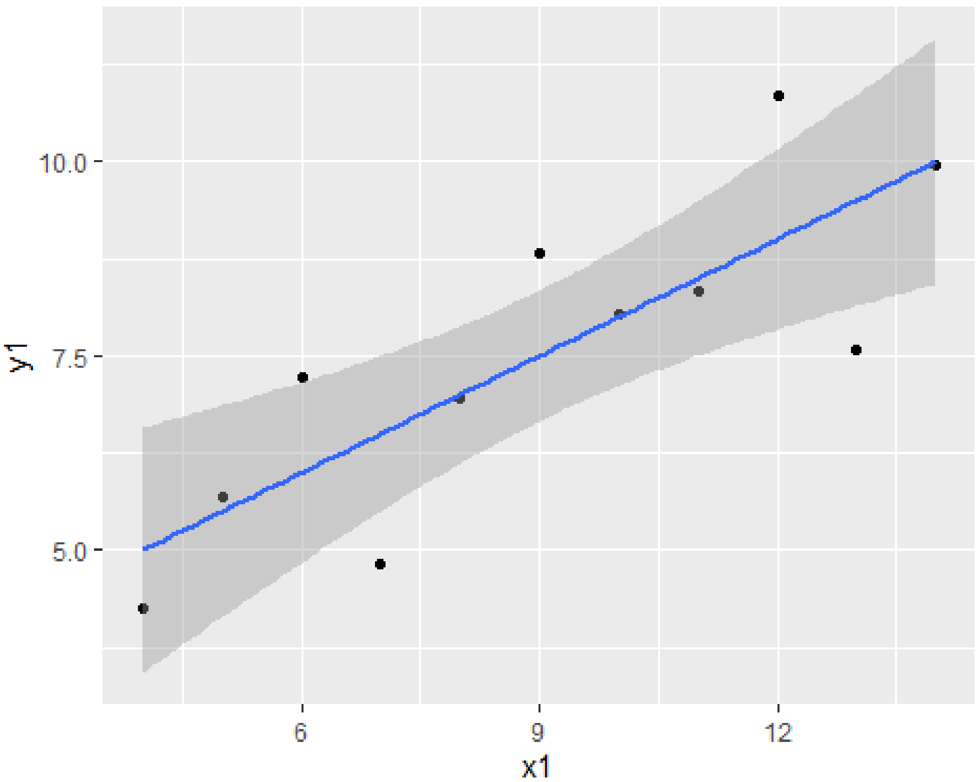
- Non-Normality
  - Histogram
  - Normal probability plot
  - Q-Q plot
  - Skewness
  - Kurtosis
  - Shapiro-Wilk test
  - Kolmogorov-Shapiro test

CASE	X1	X2	X3	X4	Y1	Y2	Y3	Y4
1	10	10	10	8	8.04	9.14	7.46	6.58
2	8	8	8	8	6.95	8.14	6.77	5.76
3	13	13	13	8	7.58	8.74	12.74	7.71
4	9	9	9	8	8.81	8.77	7.11	8.84
5	11	11	11	8	8.33	9.26	7.81	8.47
6	14	14	14	8	9.96	8.1	8.84	7.04
7	6	6	6	8	7.24	6.13	6.08	5.25
8	4	4	4	19	4.26	3.1	5.39	12.5
9	12	12	12	8	10.84	9.13	8.15	5.56
10	7	7	7	8	4.82	7.26	6.42	7.91
11	5	5	5	8	5.68	4.74	5.73	6.89

Example 2
Scatterplot 1

library(ggplot2)
ggplot(
  data = anscombe,
  mapping = **aes**(x = x1, y = y1))
+
  geom_point() **+**
  geom_smooth(method = "lm")

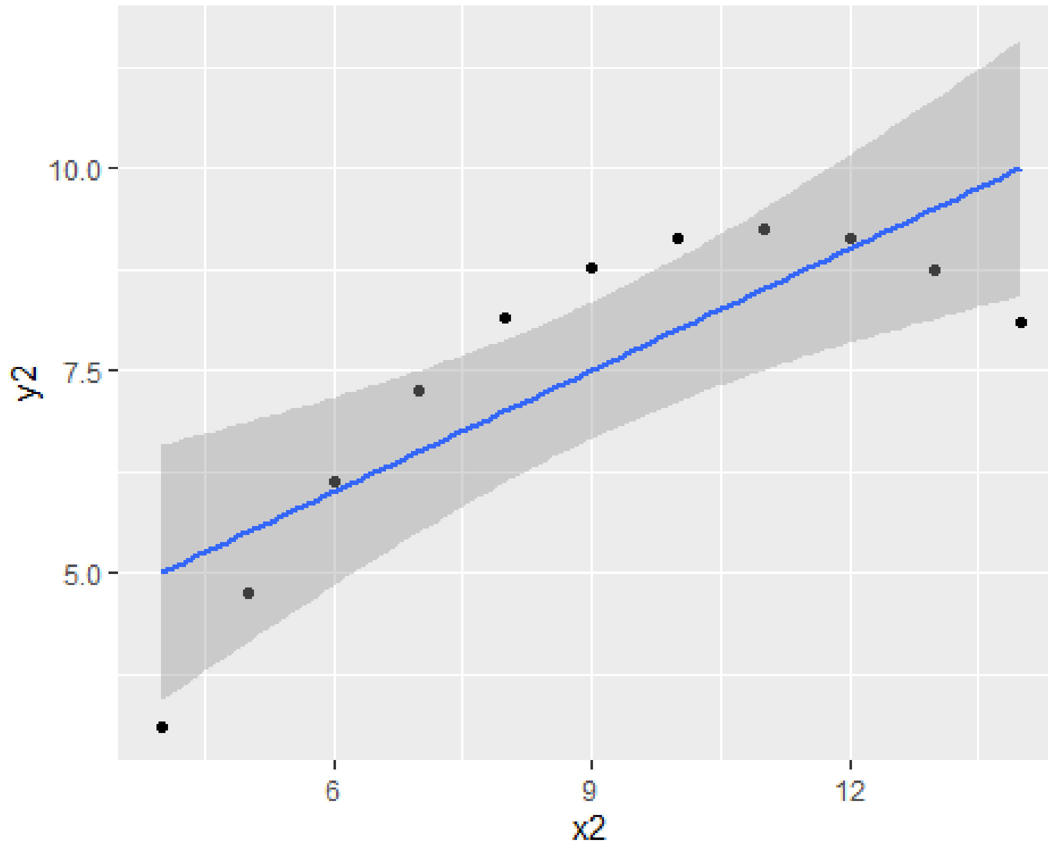
`geom_smooth()` using formula = 'y ~ x'



Example 2
Scatterplot 2

```
ggplot(
  data = anscombe,
  mapping = aes(x = x2
, y = y2)) +
  geom_point() +
  geom_smooth(method =
"lm")
```

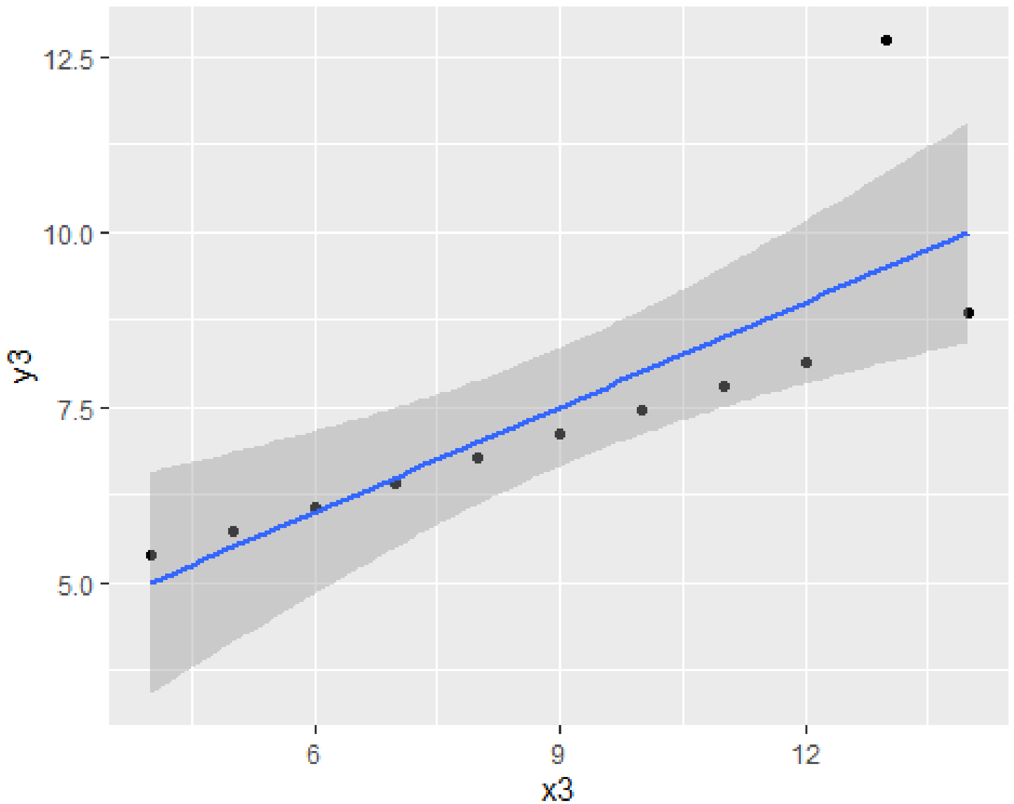
```
## `geom_smooth()` usi
ng formula = 'y ~ x'
```



Example 2
Scatterplot 3

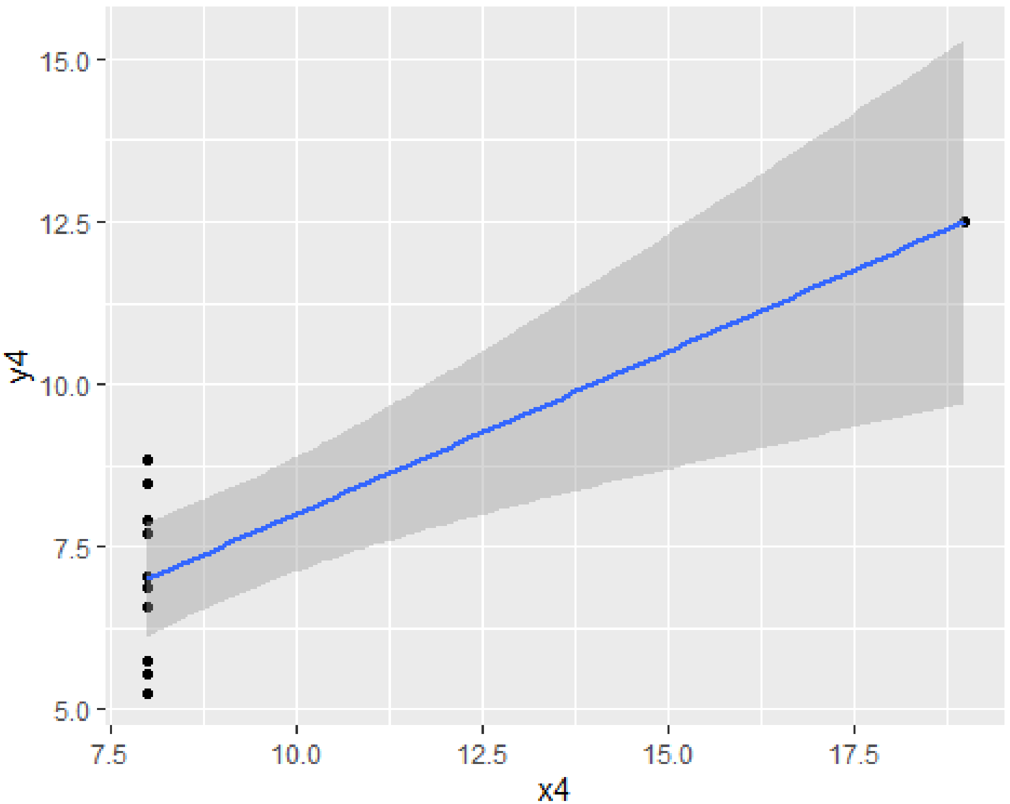
```
ggplot(
  data = anscombe,
  mapping = aes(x = x3,
y = y3)) +
  geom_point() +
  geom_smooth(method =
"lm")

## `geom_smooth()` usin
g formula = 'y ~ x'
```



Example 2
Scatterplot 4

```
ggplot(
  data = anscombe,
  mapping = aes(x = x4,
y = y4)) +
  geom_point() +
  geom_smooth(method =
"lm")
## `geom_smooth()` usin
g formula = 'y ~ x'
```



Example 2 –SLR1

```
output1 <- lm(y1~x1, anscombe)
summary(output1)
```

```
##
## Call:
## lm(formula = y1 ~ x1, data = anscombe)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.92127 -0.45577 -0.04136  0.70941  1.83882
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.0001     1.1247   2.667  0.02573 *
## x1            0.5001     0.1179   4.241  0.00217 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.237 on 9 degrees of freedom
## Multiple R-squared:  0.6665, Adjusted R-squared:  0.6295
## F-statistic: 17.99 on 1 and 9 DF,  p-value: 0.00217
```

Example 2 – SLR2

```
output2 <- lm(y2~x2, anscombe)
summary(output2)
```

```
## Call:
## lm(formula = y2 ~ x2, data = anscombe)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9009 -0.7609  0.1291  0.9491  1.2691
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.001      1.125   2.667  0.02576 *
## x2             0.500      0.118   4.239  0.00218 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.237 on 9 degrees of freedom
## Multiple R-squared:  0.6662, Adjusted R-squared:  0.6292
## F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002179
```

Example 2 – SLR 3

```
output3 <- lm(y3~x3, anscombe)
summary(output3)
```

```
##
## ## Call:
## lm(formula = y3 ~ x3, data = anscombe)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1586 -0.6146 -0.2303  0.1540  3.2411
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.0025     1.1245   2.670  0.02562 *
## x3            0.4997     0.1179   4.239  0.00218 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.236 on 9 degrees of freedom
## Multiple R-squared:  0.6663, Adjusted R-squared:  0.6292
## F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002176
```

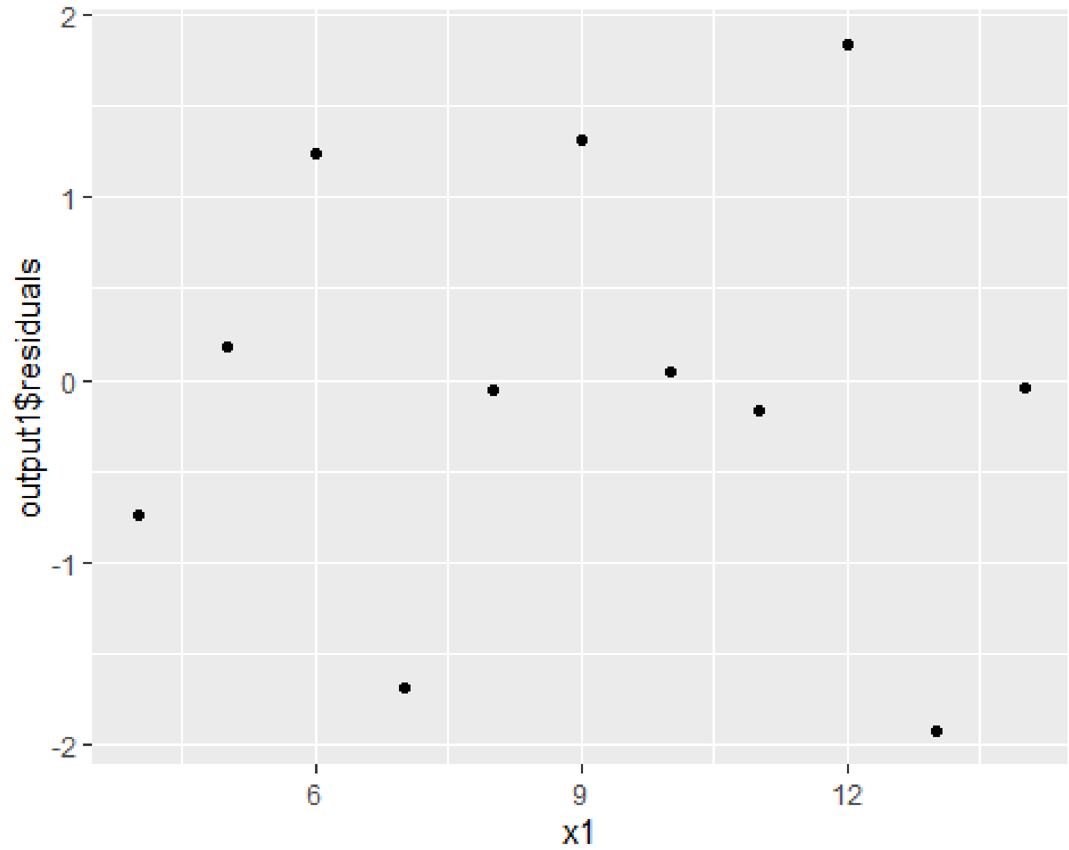
Example 2 – SLR 4

```
output4 <- lm(y4~x4, anscombe)
summary(output4)
```

```
##
Call:
## lm(formula = y4 ~ x4, data = anscombe)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -1.751 -0.831  0.000  0.809  1.839
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.0017     1.1239   2.671  0.02559 *
## x4            0.4999     0.1178   4.243  0.00216 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.236 on 9 degrees of freedom
## Multiple R-squared:  0.6667, Adjusted R-squared:  0.6297
## F-statistic:    18 on 1 and 9 DF,  p-value: 0.002165
```

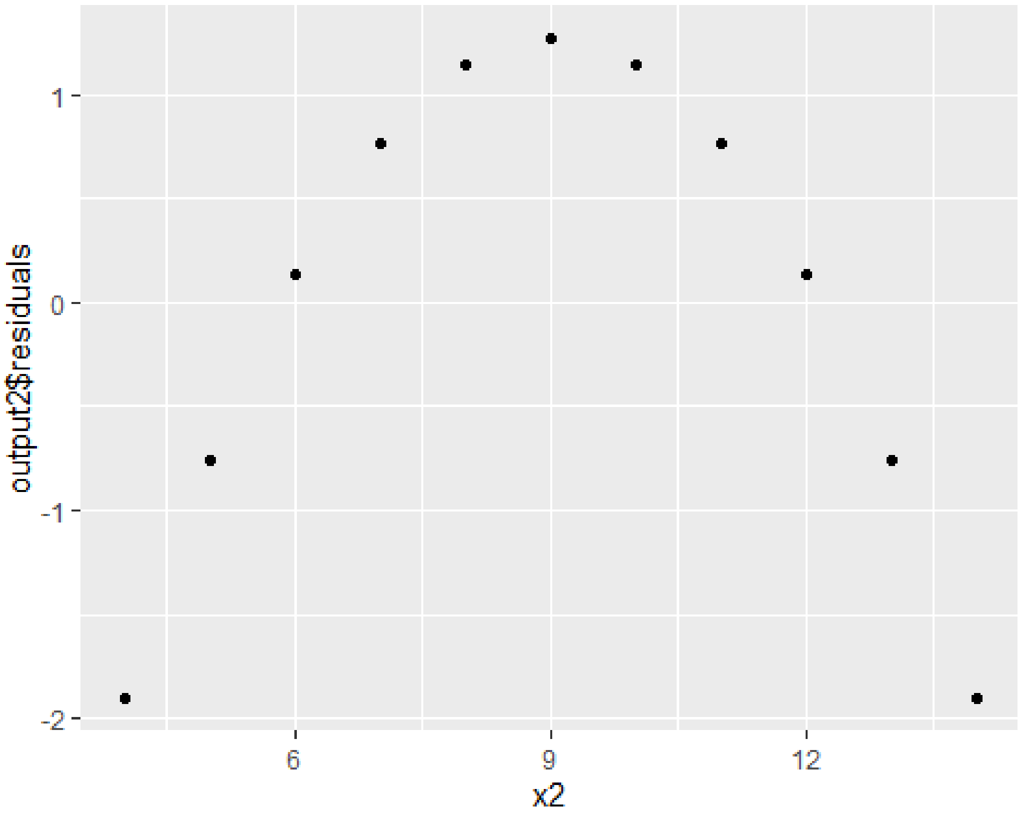
Example 2
Residual plot 1

```
ggplot(
  data = output1,
  mapping = aes(x = x1
, y = output1$residual
s)) +
  geom_point()
```



Example 2
Residual plot 2

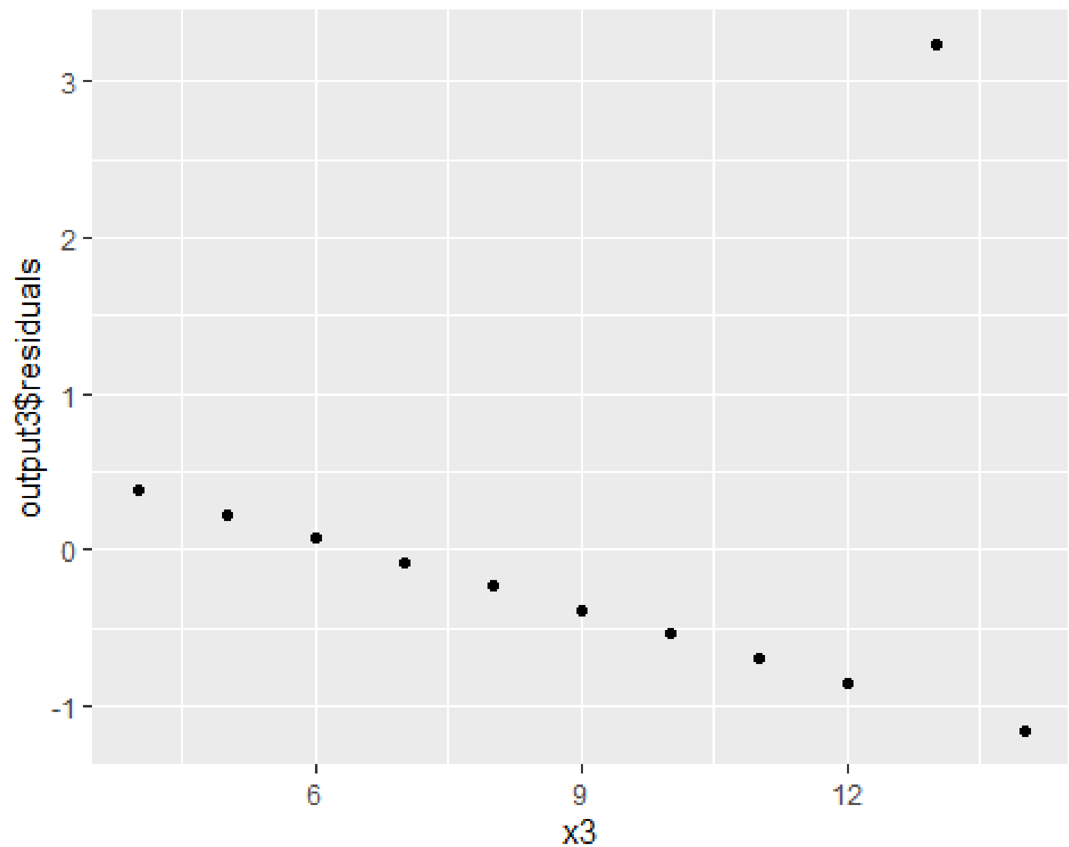
```
ggplot(
  data = output2,
  mapping = aes(x = x2
, y = output2$residual
s)) +
 geom_point()
```



Example 2

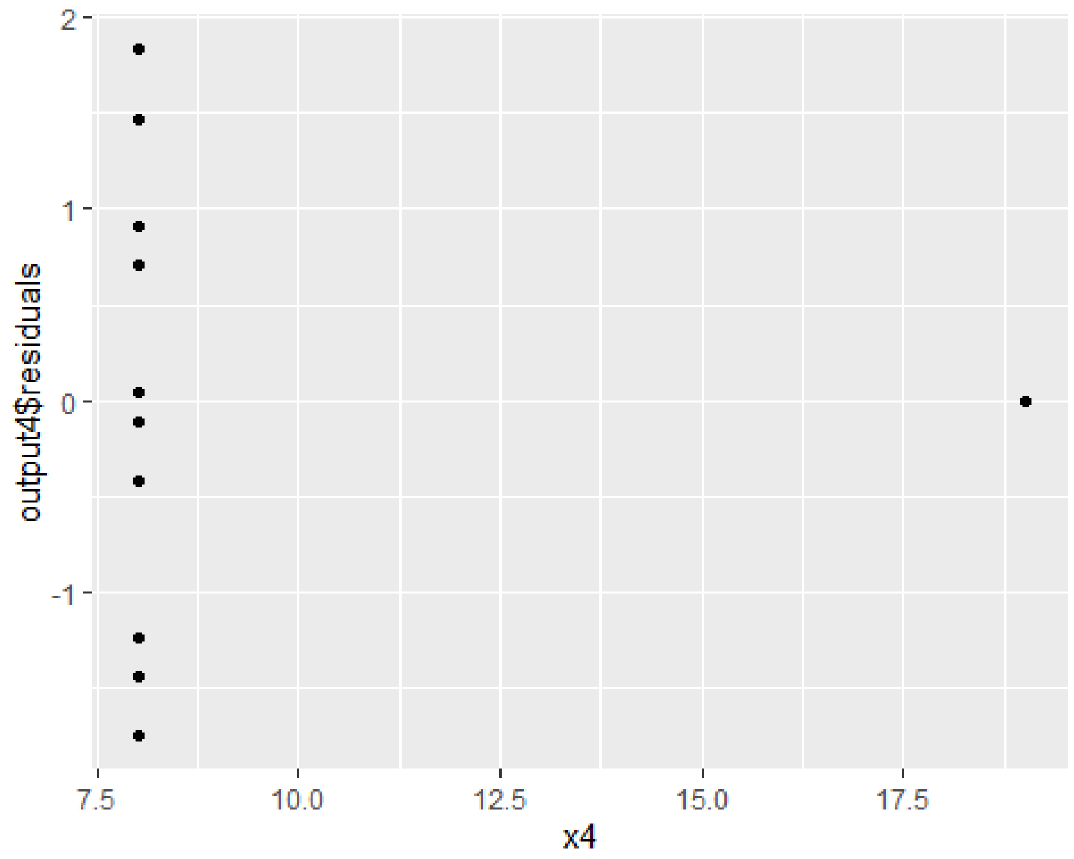
Residual plot 3

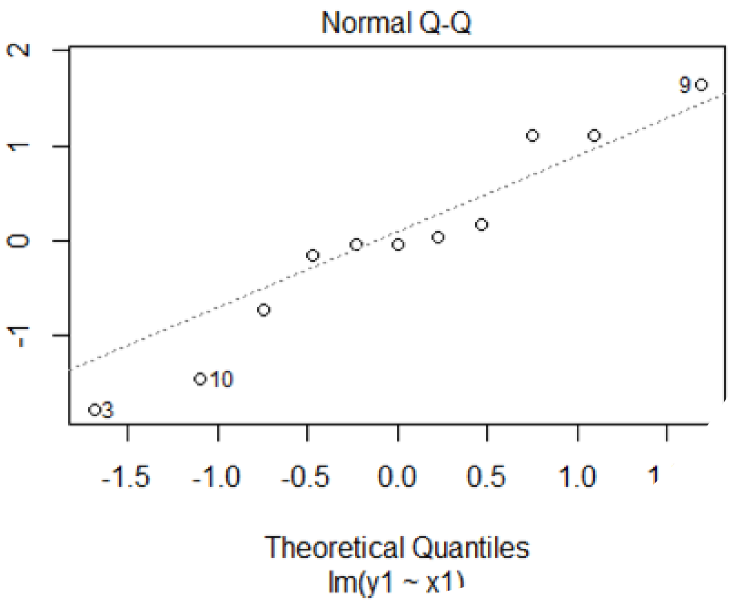
```
ggplot(
  data = output3,
  mapping = aes(x = x3
, y = output3$residual
s)) +
  geom_point()
```



Example 2
Residual plot 4

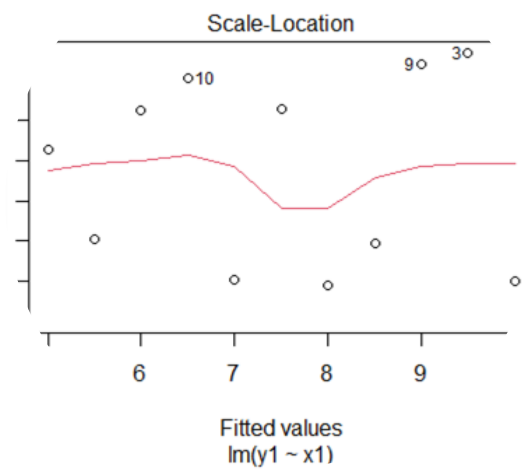
```
ggplot(
  data = output4,
  mapping = aes(x = x4
, y = output4$residual
s)) +
  geom_point()
```

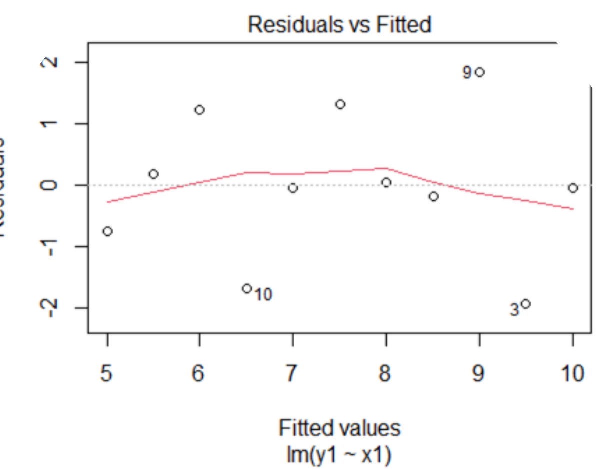


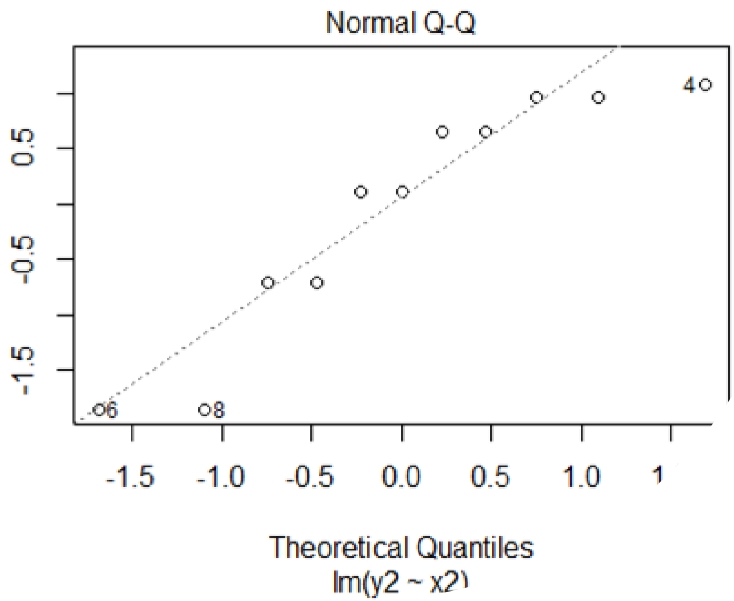


Example 2
Normal Probability Plot

`plot(output1)`

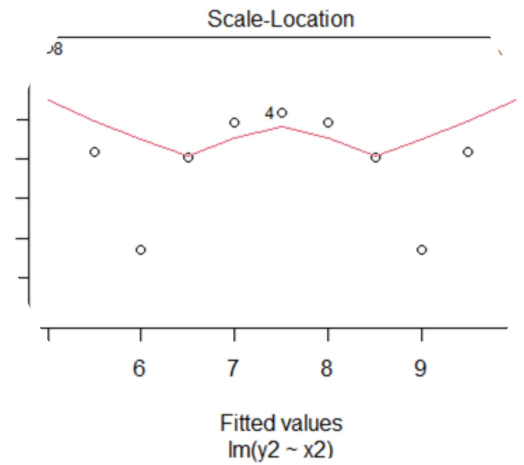


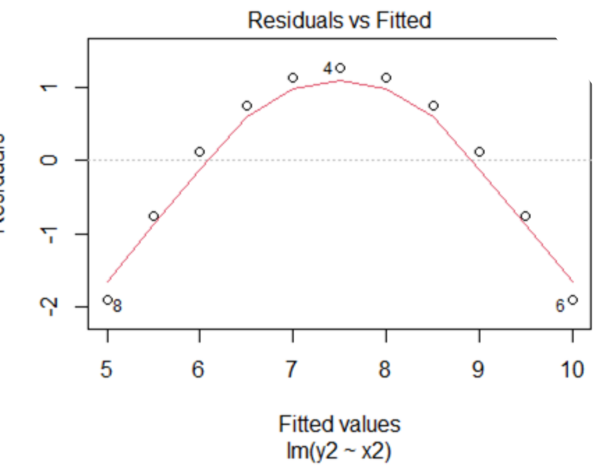


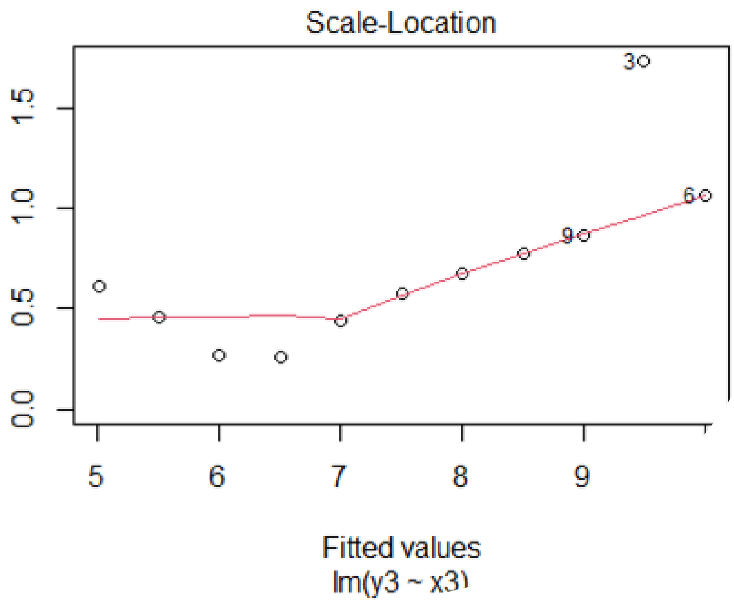


Example 2
Normal Probability Plot

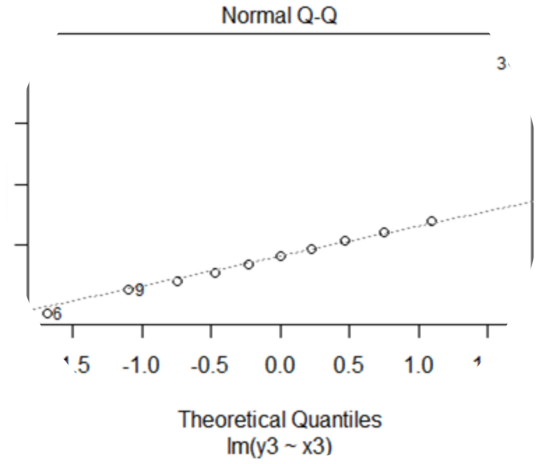
`plot(output2)`



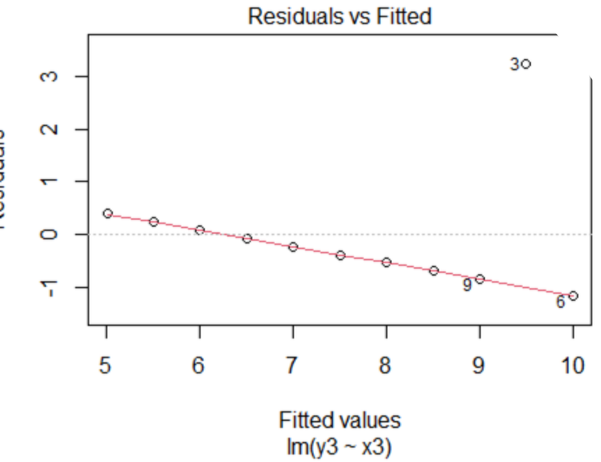


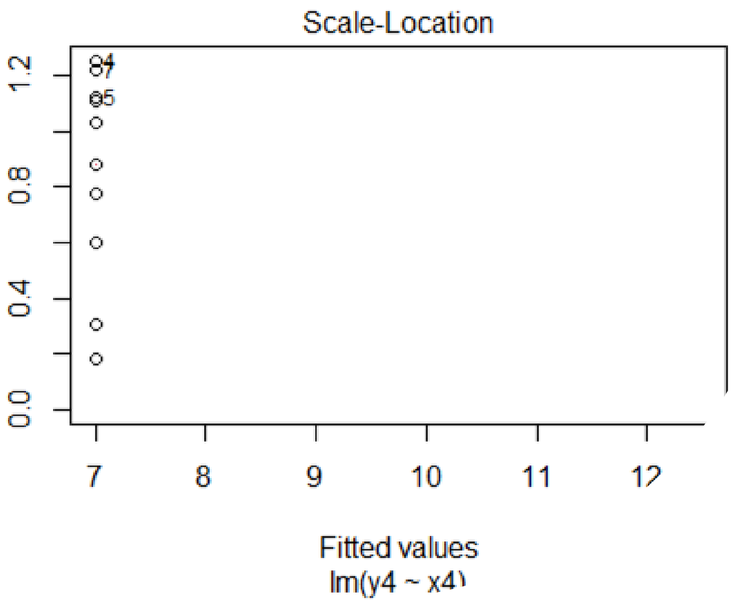


Example 2
Normal Probability Plot

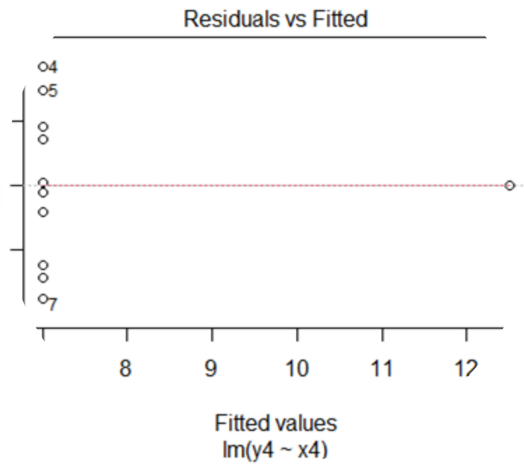


`plot(output3)`

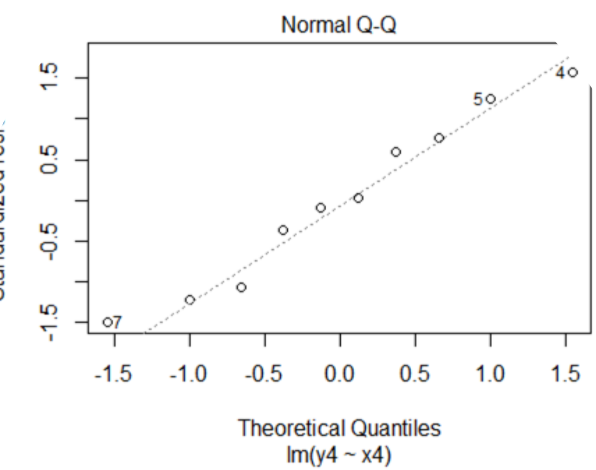




Example 2
Normal Probability Plot




```
plot(output4)
```



Remedial Measures in SLR

- Instead of SLR model, develop and use a more appropriate model
  - Complex model but better
  - Complex procedures for estimating parameters
- Use Transformations on the Data
  - Simpler methods of estimation
  - Fewer parameters than complex models

Remedial Measures in SLR
Nonlinearity of Regression Function

- Fit a quadratic regression function
- Fit an exponential regression function
- Transformations
  - Error distribution is approximately Normal
  - Error terms have close to constant variance
  - Transform only X variable
  - Transforming Y variable will change the shape of the distribution
  - Prototype transformations

Remedial Measures in SLR
Nonindependence of error terms

- Use model that calls for correlated error terms
  - Use first difference
  - Model for Time Series data

Remedial Measures in SLR
Nonconstancy of error variance

- If error terms vary in a systematic manner, use method of weighted least squares to obtain the estimate of the parameters
- Transformations on Y
  - Simultaneous transformation on X
  - Scatterplots and residual p[lots should be looked at

Remedial Measures in SLR
Nonnormality of error terms

- Often the same transformation that stabilizes the variance, helps in approximately normalizing the error terms
- Utilize variance stabilizing transformation first
  - Check residuals for departure from normality

Remedial Measures in SLR
Outlying observations

- Least squares method leads to serious distortions in the estimated regression function
- Can’t discard the outlying values
- Robust estimation procedures

Example 3: Predict Price of Dinner

Zagat Survey 2001: New York City Restaurants , Zagat, New York

- New Italian Restaurant
- Y (Price): the price (in $US) of dinner (including 1 drink & a tip)
- x_1 (Food): customer rating of the food (out of 30)
- x_2 (Décor): customer rating of the decor (out of 30)
- x_3 (Service): customer rating of the service (out of 30)
- x_4 (East) dummy variable = 1 (0) if the restaurant is east (west) of Fifth Avenue

Example 3: Predict Price of Dinner

Zagat Survey 2001: New York City Restaurants , Zagat, New York

- Develop a regression model that directly predicts the price of dinner (in dollars) using a subset or all of the 4 potential predictor variables listed above.
- Determine which of the predictor variables Food, Décor and Service has the largest estimated effect on Price? Is this effect also the most statistically significant?
- If the aim is to choose the location of the restaurant so that the price achieved for dinner is maximized, should the new restaurant be on the east or west of Fifth Avenue?
- Does it seem possible to achieve a price premium for “setting a new standard for high-quality service in Manhattan” for Italian restaurants?

Example 3: Predict Price of Dinner

Zagat Survey 2001: New York City Restaurants , Zagat, New York

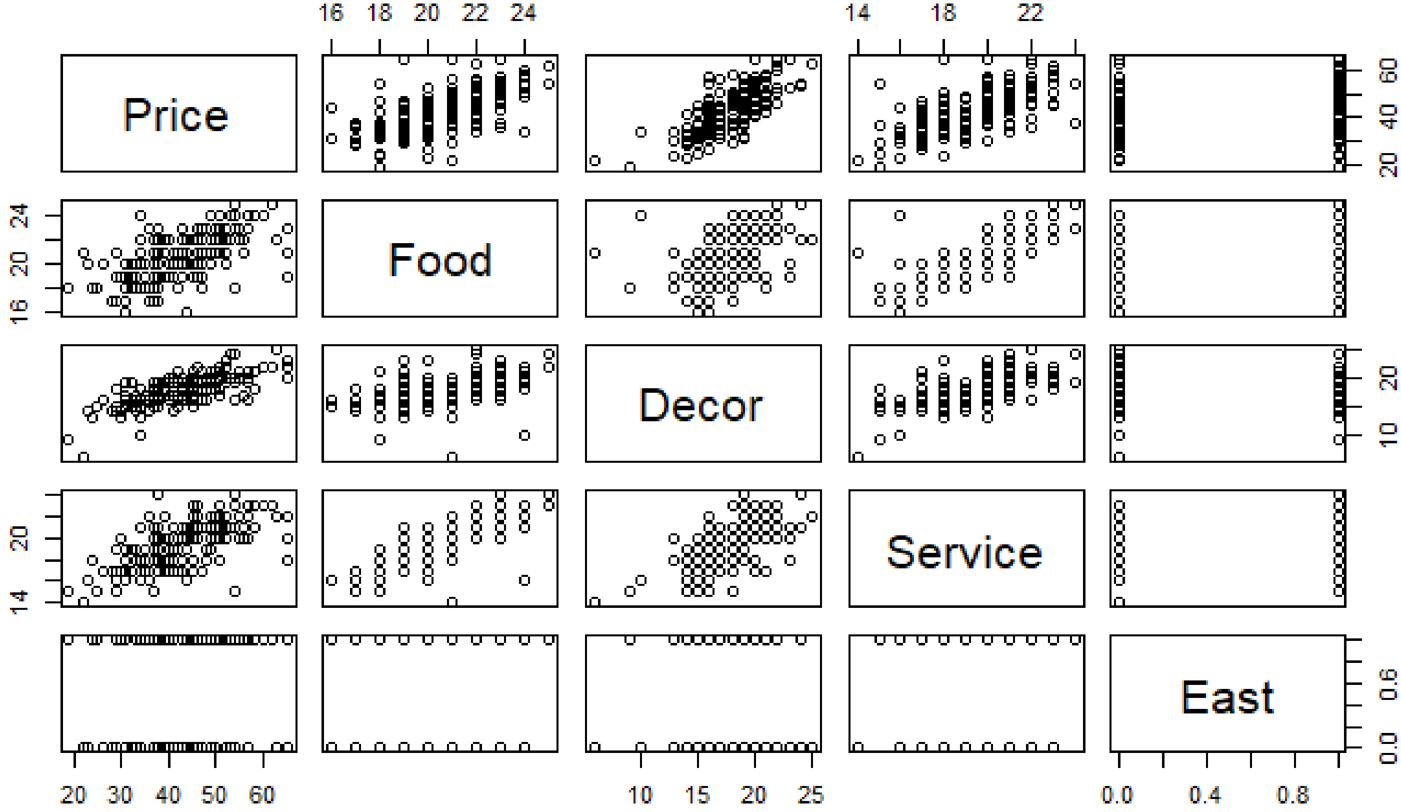
Model

$$\text{Price} = \beta_0 + \beta_1 \text{(Food)} + \beta_2 \text{(Décor)} + \beta_3 \text{(Service)} + \beta_4 \text{(East)} + \varepsilon$$

<dbl>	Restaurant <chr>	Price <dbl>	Food <dbl>	Decor <dbl>	Service <dbl>	East <dbl>
1	Daniella Ristorante	43	22	18	20	0
2	Tello's Ristorante	32	20	19	19	0
3	Biricchino	34	21	13	18	0
4	Bottino	41	20	20	17	0
5	Da Umberto	54	24	19	21	0
6	Le Madri	52	22	22	21	0

Example 3: Predict Price of Dinner

Correlation Plots



Correlation

- `cor(data_restau)`
- ```
  ##                 Price      Food     Decor   Service
  ## Price   1.0000000 0.6270435 0.7243525 0.6411402
  ## Food    0.6270435 1.0000000 0.5039161 0.7945248
  ## Decor   0.7243525 0.5039161 1.0000000 0.6453306
  ## Service 0.6411402 0.7945248 0.6453306 1.0000000
  ```
  ```

Coefficients:

(Intercept)	Food	Decor	Service	East
-24.023800	1.538120	1.910087	-0.002727	2.068050

Residuals:

Min	1Q	Median	3Q	Max
-14.0465	-3.8837	0.0373	3.3942	17.7491

Coefficients:

	Estimate	Std. Error	t value	Pr(>|t|)	
(Intercept)	-24.023800	4.708359	-5.102	9.24e-07	***
Food	1.538120	0.368951	4.169	4.96e-05	***
Decor	1.910087	0.217005	8.802	1.87e-15	***
Service	-0.002727	0.396232	-0.007	0.9945	
East	2.068050	0.946739	2.184	0.0304	*

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.738 on 163 degrees of freedom

Multiple R-squared: 0.6279, Adjusted R-squared: 0.6187

F-statistic: 68.76 on 4 and 163 DF, p-value: < 2.2e-16

- Testing the validity of the model
  - We pose the question:

    Is there at least one independent variable linearly related to the dependent variable?
  - To answer the question we test the hypothesis

    H_0: $\beta_1 = \beta_2 = \ldots = \beta_k = 0$

    H_1: At least one β_i is not equal to zero.
  - If at least one β_i is not equal to zero, the model is valid.

Example 3

ANOVA results

Analysis of Variance Table

Response: Price

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Food	1	5670.3	5670.3	172.2269	< 2e-16 ***
Decor	1	3223.7	3223.7	97.9142	< 2e-16 ***
Service	1	3.9	3.9	0.1192	0.73037
East	1	157.1	157.1	4.7716	0.03036 *
Residuals	163	5366.5	32.9		